

Q. Consider any data of 7 bits and identify respective 5 bits parity. Apply the hamming code with even parity check method to generate the codeword. Also identify the position of corrupted bit.

P1	P2	1	P4	0	1	1	P8	0	1	0
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Data $\rightarrow$ 1011010

$$P1 = 1 \oplus 0 \oplus 1 \oplus 0 \oplus 0 = 0$$

$$P2 = 1 \oplus 1 \oplus 1 \oplus 1 \oplus 0 = 0$$

~~P4~~

$$P4 = 0 \oplus 1 \oplus 1 = 0$$

$$P8 = 0 \oplus 1 \oplus 0 = 1$$

Corrupted Codeword: 00100101010
 Original Codeword: 00100111010

This bit is corrupted and needs to be flipped

Checking $P1 = 0 \oplus 1 \oplus 0 \oplus 0 \oplus 0 \oplus 0 \oplus 0 = 1$ (Failed)

Checking $P2 = 0 \oplus 1 \oplus 1 \oplus 0 \oplus 1 \oplus 0 = 1$ (Failed)

Checking $P4 = 0 \oplus 0 \oplus 1 \oplus 0 = 1$ (Failed)

Checking $P8 = 1 \oplus 0 \oplus 1 \oplus 0 = 0$ (Passed)

Syndrome ($P8, P4, P2, P1$) = 0111

Converting Binary Syndrome to decimal = $0 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$
 $= 4 + 2 + 1$
 $= 7$

Thus, 7 pinpoints the exact position of the corrupted bit.